From Limit Order Book Data to Hawkes Processes:
A Hands-on Introduction to Self-Exciting Price Dynamics

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Motivation

Training RL strategies on financial data can be done either offline (historical replay) or using simulators.

Each security provides only **one realized trajectory** of its LOB dynamics. If a strategy is trained on this single path:

- It risks severe **overfitting**
- Apparent in-sample performance may not generalize
- True out-of-sample performance can collapse

Adding more data would help — but we do *not* know the true data-generating process. This is where **synthetic data via simulators** becomes useful.

However, pure simulation introduces bias. This can be mitigated by:

1. Building simulators that replicate key **stylized facts** (clustering, heavy tails, spread dynamics, impact)
2. Combining simulated data with **real out-of-sample testing**

To train RL agents, we must first model and simulate the underlying market microstructure realistically.

Outline

1. Part I: Challenging the Efficient Market Hypothesis
2. Part II: Limit Order Book Mechanics
3. Part III: Hawkes Process Theory
4. Part IV: Hands-on Project
5. Part V: Compound Hawkes Process for LOB
6. Part VI: Calibration and Estimation
7. Part VII: Empirical Results
8. Conclusion

The Standard View: Efficient Markets

The Efficient Market Hypothesis (EMH) tells us:

- Prices fully reflect all available information
- Price changes are unpredictable (for e.g. random walk)
- Past prices contain no information about future prices
- Trading strategies cannot consistently beat the market

Mathematical formulation:

$$P_{t+1} = P_t + \epsilon_{t+1}$$
$$\mathbb{E}[\epsilon_{t+1}|\mathcal{F}_t] = 0$$

where $\mathcal{F}_t$ is all information available at time t.

Question: Is this really what we observe in high-frequency data?

Reality Check: What Actually Happens in Markets

Stylized Facts That Challenge EMH:

1. **Volatility Clustering:**
   - Mandelbrot (1963): "Large changes tend to be followed by large changes"
   - Violates independence assumption
2. **Heavy-Tailed Returns:** Fat tails in return distributions
   - More extreme events than Gaussian would predict
   - $P(|r| > x) \sim x^{-\alpha}$ with $\alpha \approx 3 - 4$
3. **Microstructure Effects:** Order flow predicts short-term prices
   - Kyle (1985): Informed trading moves prices
   - Trade direction has persistent impact

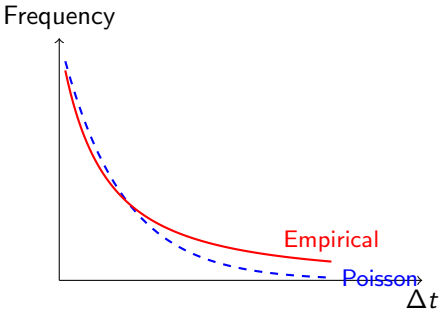
The High-Frequency Reality: Orders Cluster

Empirical Observation:

- Market orders arrive in clusters
- After a trade, more trades follow
- Not just "news arrival" – endogenous feedback

Inter-arrival times:

- Heavy left tail (very short times)
- Heavy right tail (long quiet periods)
- Cannot be Poisson!



Inter-arrival time distribution

This suggests events are *self-exciting*: past events trigger future events

Market Impact: Prices Move When You Trade

Key Finding: Trading causes predictable price movements

- Buy 10,000 shares $\Rightarrow$ price increases
- Sell 10,000 shares $\Rightarrow$ price decreases
- Impact is **concave** in order size: $\Delta P \propto Q^{\alpha}$, $\alpha < 1$
- Impact is **temporary and permanent**:
  - Immediate impact from liquidity consumption
  - Persistent impact from information revelation

Mathematical Framework (Cont et al. 2014):

$$\mathbb{E}[\Delta P | Q] = f(Q) = \sigma \cdot Q^{0.5-0.7}$$

If markets were truly efficient, there should be no predictable impact!

Why Traditional Models Fail

Property	EMH/Geometric BM	Reality
Return autocorrelation	Zero	Negative at high-freq
Volatility clustering	No	Yes
Heavy tails	No (Gaussian)	Yes (power law)
Market impact	Linear	Concave in size
Order clustering	No	Yes
Price discreteness	Ignored	Important

Fundamental issue: EMH assumes:

1. Exogenous information arrival (Poisson)
2. Instantaneous adjustment to information
3. No strategic interaction

We need models that capture *endogenous dynamics* and *feedback effects*

What is a Limit Order Book?

Modern markets are order-driven:

- No designated market maker
- All participants can provide liquidity
- Orders are displayed in a **limit order book (LOB)**

Example LOB snapshot:

Bid		Ask	
Qty	Price	Price	Qty
500	149.98	150.00	300
800	149.97	150.01	600
400	149.96	150.02	200

Key concepts:

- **Best bid:** $149.98
- **Best ask:** $150.00
- **Spread:** 2 cents (2 ticks)
- **Mid-price:** $149.99

Price formation happens through order flow dynamics!

Three Types of Orders

1. **Limit Orders (LO):** Post liquidity
   - Specify price and quantity
   - Wait in book until matched or cancelled
   - Example: Buy 100 @ $149.97
2. **Market Orders (MO):** Consume liquidity
   - Execute immediately at best available price
   - Remove liquidity from book
   - Example: Buy 300 shares (at $150.00)
3. **Cancellations (CO):** Remove posted orders
   - Cancel all or part of a limit order
   - 99.3% are full cancellations (empirical observation)
   - Example: Cancel buy order at $149.97

Price changes when best bid/ask changes!

Order Flow Statistics: Stylized Facts

From Apple Inc. (AAPL) NASDAQ data (2019):

Order sizes:

- 40% of market orders are 100 shares
- 60% of limit orders are 100 shares
- Strong preference for round numbers
- Heavy right tail (large orders exist)

Order intensity:

- U-shaped intraday pattern
- High at open and close
- Low at midday

These facts motivate our modelling choices!

Spread dynamics:

- 97.4% of in-spread orders are 1 tick from best
- 97.5% of price changes are 1 tick
- Spread is mean-reverting
- Average spread: 1.7 ticks for AAPL

Price changes:

- Caused by: queue depletion or in-spread orders
- Clustering behavior
- Non-Gaussian returns

Queue Dynamics and Price Formation

How prices change:

1. **Queue Depletion (QD):** Best bid/ask fully consumed
   - Large market order eats through level
   - Or: all limit orders at level cancelled
   - $\Rightarrow$ Spread widens, price jumps
2. **In-Spread Orders (IS):** Limit order inside spread
   - New best bid/ask created
   - $\Rightarrow$ Spread narrows, price moves

Mathematical representation:

$$P(t^+) - P(t) = -\frac{\eta}{2}\xi\, dN^{(IS)}(t) + \frac{\eta}{2}\xi\, dN^{(QD)}(t)$$

where η = tick size, $\xi = +1$ for ask side, -1 for bid side

Break

Questions so far?

Point Processes: Basic Definitions

Point Process: A random collection of points in time

Counting Process: $N(t)$ = number of events in $[0, t]$

- $N(0) = 0$
- $N(t)$ is non-decreasing
- $N(t)$ has right-continuous paths with left limits

Intensity Process: $\lambda(t)$ = instantaneous arrival rate

$$\mathbb{P}(N(t + dt) - N(t) = 1) = \lambda(t)\, dt + o(dt)$$

Key Property:

$$\mathbb{E}[dN(t)|\mathcal{F}_{t^-}] = \lambda(t)\, dt$$

where $\mathcal{F}_{t^-}$ is the history up to time t^-

Poisson Process: The Baseline

Homogeneous Poisson Process:

$$\lambda(t) = \mu \quad \text{(constant)}$$

Properties:

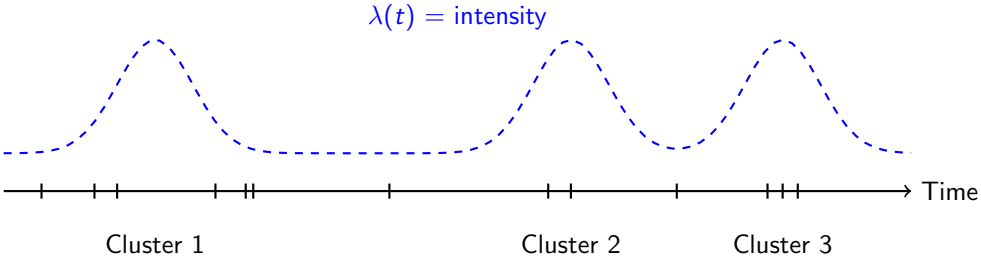
- Inter-arrival times are i.i.d. $\text{Exp}(\mu)$
- Number of events in $[t, t+s]$ is $\text{Poisson}(\mu s)$
- Events are *independent* – no clustering

Why Poisson fails for financial data:

1. Cannot generate volatility clustering
2. Cannot produce heavy-tailed inter-arrival times
3. No correlation in event counts
4. No feedback or excitation

We need intensity to depend on history: $\lambda(t|\mathcal{F}_{t^-})$

Enter Hawkes Process: A Different Paradigm



Key idea: Event arrival rate $\lambda(t)$ depends on *past events*

$$\lambda(t) = \underbrace{\mu}_{\text{baseline}} + \sum_{t_i < t} \underbrace{\phi(t - t_i)}_{\text{excitation from past}}$$

This is a **Hawkes process** – the foundation of our approach

Hawkes Process: Self-Exciting Dynamics

Definition (Hawkes, 1971): A point process with intensity

$$\lambda(t) = \mu + \sum_{t_i < t} \phi(t - t_i)$$

where:

- $\mu > 0$: exogenous (baseline) intensity
- $\phi(s) \geq 0$: excitation kernel (memory function)
- $t_1, t_2, \ldots$: past event times

Intuition:

- Each event increases intensity by $\phi(\cdot)$
- Effect decays over time
- Past events make future events more likely
- This is *self-excitation*

Key difference from Poisson: $\lambda(t)$ is stochastic and path-dependent

Univariate Hawkes: Integral Formulation

Equivalent representation:

$$\lambda(t) = \mu + \int_0^t \phi(t-s)\, dN(s)$$

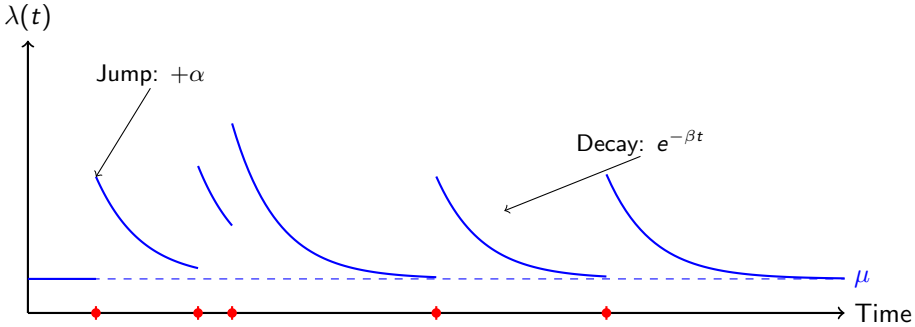
Discretized version:

$$\lambda(t) = \mu + \sum_{i:t_i<t} \phi(t-t_i)$$

Common kernel choices:

1. **Exponential:** $\phi(t) = \alpha e^{-\beta t}$
   - Memoryless decay
   - One parameter for strength (α), one for decay (β)
2. **Power law:** $\phi(t) = \alpha(1+\gamma t)^{-\beta}$
   - Long memory (if $\beta \leq 1$)
   - Better fit for financial data

Example: Exponential Kernel Dynamics



Parameters: $\mu = 0.5$, $\alpha = 1.5$, $\beta = 1.5$

After each event: intensity jumps by α, then decays exponentially

Multivariate Hawkes Process

For d event types: Vector process $(N^{(1)}(t), \ldots, N^{(d)}(t))$

$$\lambda^{(i)}(t) = \mu^{(i)} + \sum_{j=1}^{d} \sum_{t_j^k < t} \phi^{(j \to i)}(t - t_j^k)$$

where $t_j^1, t_j^2, \ldots$ are event times of type j

Interpretation:

- $\phi^{(j \to i)}$: impact of event type j on intensity of type i
- **Self-excitation:** $\phi^{(i \to i)} > 0$
- **Cross-excitation:** $\phi^{(j \to i)} > 0$ for $j \neq i$
- **Inhibition:** $\phi^{(j \to i)} < 0$ (damping effect)

Matrix notation:

$$\boldsymbol{\lambda}(t) = \boldsymbol{\mu} + \int_0^t \boldsymbol{\Phi}(t - s)\, d\mathbf{N}(s)$$

where $\boldsymbol{\Phi}$ is the $d \times d$ kernel matrix

Branching Interpretation

Hawkes process as branching process:

- **Immigrants:** Arrive at rate μ (exogenous events)
- **Offspring:** Each event generates children according to $\phi(\cdot)$
- **Branching ratio:**

$$n = \int_0^\infty \phi(s)\, ds = \text{expected children per event}$$

Mean intensity:

$$\mathbb{E}[\lambda] = \frac{\mu}{1-n}$$

Examples:

- If $n = 0.5$: Each event generates 0.5 children on average $\Rightarrow \mathbb{E}[\lambda] = 2\mu$
- If $n = 0.9$: Each event generates 0.9 children $\Rightarrow \mathbb{E}[\lambda] = 10\mu$
- If $n \geq 1$: Process explodes!

High branching ratio = strong feedback

Why Hawkes for LOB? Theoretical Motivation

1. **Captures clustering:**
   - Orders arrive in bursts
   - Volatility clustering emerges naturally
   - Heavy-tailed inter-arrival times
2. **Endogenous dynamics:**
   - No need to model external news
   - Feedback is built into the model
   - Market impact is implicit
3. **Mathematically tractable:**
   - Likelihood is computable
   - Simulation via thinning algorithm
   - Rich theoretical properties
4. **Flexible:**
   - Multivariate formulation for different event types
   - Can include inhibitory effects
   - Can be made state-dependent

Break

Questions before we move to a project?

Project Overview

Goal: Build empirical pipeline from raw LOB data to fitted Hawkes model

Data source: LOBSTER database : https://data.lobsterdata.com/info/DataSamples.php

- High-frequency limit order book data
- Message-by-message reconstruction
- Multiple stocks available

Code pipeline:

1. Load and clean LOBSTER data
2. Extract event-level dataset
3. Identify price-changing events
4. Compute stylized facts
5. Fit simple Hawkes model (1D, exponential kernel)
6. Diagnose and extend

Provided code is modular and designed for experimentation

Simplified Setup: 1D Hawkes

For this project: Focus on univariate model

Event definition: Price changes (up or down)

- Ignore direction initially
- Just model when price moves

Model:

$$\lambda(t) = \mu + \alpha \sum_{t_i < t} e^{-\beta(t - t_i)}$$

Parameters:

- μ: baseline rate of price changes
- α: jump in intensity after each event
- β: decay rate
- Branching ratio: $n = \alpha/\beta$

Much simpler than full 12D model, but captures key ideas

Step 1: From LOBSTER to Events

LOBSTER format: Two files per stock-date

1. `message` file: Event-by-event orders
   - Timestamp, event type, order ID, size, price, direction
2. `orderbook` file: LOB snapshots after each event
   - Best bid/ask prices and quantities at multiple levels

Code task:

- Parse message file
- Identify price-changing events:
  - Best bid increases
  - Best ask decreases
  - (or opposite for price down)
- Extract timestamps: $t_1, t_2, \ldots, t_n$

Provided: Python functions to load and filter LOBSTER data:
https://github.com/konqr/UoE_LOBHawkes

Step 2: Exploratory Analysis

Compute stylized facts:

1. **Inter-arrival time distribution**

$$\Delta t_i = t_{i+1} - t_i$$

   - Plot histogram on log-log scale
   - Compare to exponential (Poisson benchmark)

2. **Event count autocorrelation**
   - Bin events into fixed intervals (e.g., 1 second)
   - Compute $N_1, N_2, \ldots$ (counts per bin)
   - Plot ACF of $\{N_k\}$

3. **Signed price move correlation**
   - Define $X_i = +1$ (up) or -1 (down)
   - Test $\mathrm{Corr}(X_i, X_{i+k})$ for lags k

These establish *need* for self-exciting model

Step 3: Binned VAR Analysis

Intuition: If events are self-exciting, event counts should predict future counts

Method: Fit Vector Autoregression (VAR)

$$N_t = c + \sum_{k=1}^{p} A_k N_{t-k} + \epsilon_t$$

For 1D case:

$$N_t = c + \sum_{k=1}^{p} a_k N_{t-k} + \epsilon_t$$

Code analysis:

- Vary bin size: $\Delta t \in \{0.1, 0.5, 1, 5, 10\}$ seconds
- Fit VAR(p) for $p \in \{1, 2, 3, 5\}$
- Test: Are coefficients a_k significant?
- Conclusion: Evidence of temporal dependence $\Rightarrow$ Hawkes appropriate

Step 4: Fitting 1D Hawkes (Exponential Kernel)

Model:

$$\lambda(t) = \mu + \alpha \sum_{t_i < t} e^{-\beta(t - t_i)}$$

Log-likelihood:

$$\ell(\mu, \alpha, \beta) = -\int_0^T \lambda(t)\, dt + \sum_{i=1}^{n} \log \lambda(t_i)$$

Efficient recursion:

$$R_i = e^{-\beta(t_i - t_{i-1})}(1 + R_{i-1})$$
$$\lambda(t_i) = \mu + \alpha R_i$$

Optimization:

- Use `scipy.optimize` or similar
- Initialize: $\mu = \bar{\lambda}/2$, $\alpha = \bar{\lambda}/4$, $\beta = 1$
- Constraint: $\alpha/\beta < 1$ (stability)

Provided: Python implementation of likelihood and optimization

Step 5: Model Diagnostics

After fitting: Check model quality

1. **Residual analysis:**
   - Transformed times: $\tau_i = \int_0^{t_i} \lambda(s)\, ds$
   - Should be unit-rate Poisson if model correct
   - Plot: $\tau_{i+1} - \tau_i$ vs Exp(1)
2. **Branching ratio:**

$$n = \frac{\alpha}{\beta}$$

   - Interpretation: Expected children per event
   - Typical values: $n \in [0.3, 0.9]$ for financial data
3. **Simulation:**
   - Use thinning algorithm
   - Compare simulated stylized facts to empirical

Suggested Extensions for Students

Choose one of these to explore:

1. **Event definition:**
   - Separate up/down price moves (2D Hawkes)
   - Include market order events
   - Test impact on results
2. **Time aggregation:**
   - Vary bin size in VAR analysis
   - How does memory structure change?
3. **Kernel comparison:**
   - Fit power-law kernel: $\phi(t) = \alpha(1+t)^{-\beta}$
   - Compare AIC with exponential
4. **Different stocks:**
   - Compare large-tick vs small-tick stocks
   - How do parameters differ?

Break

Question?

Limitation: Constant Order Sizes

Standard Hawkes models assume:

- Each event has unit effect
- Order sizes are constant (often normalized to 1)

Reality:

- Order sizes vary enormously
- Strong preference for round numbers (100, 500, 1000 shares)
- Queue depletion depends on order size
- Price impact depends on order size

Example:

- Market order for 50 shares: Doesn't deplete queue (300 shares available)
- Market order for 500 shares: Depletes queue $\Rightarrow$ price moves

We need to model order sizes explicitly!

Compound Point Processes

Definition: A compound point process (CPP) is

$$Z(t) = \sum_{i=1}^{N(t)} Y_i$$

where:

- $N(t)$: counting process (number of events)
- Y_i: i.i.d. random variables (jump sizes)
- $Y_i \sim P_\Theta(\cdot)$: jump size distribution with parameters Θ

Compound Hawkes Process (CHP):

- $N(t)$ is a Hawkes process
- Each event has a random size Y_i

Key assumption:

Order sizes are independent of past event times

Evidence: Hoeffding independence test statistic $= 0.00068$ (very small)

Order Size Distributions: Empirical Evidence

Apple Inc. (AAPL) - 2 weeks of data:

Market Orders:

- Modal size: 100 shares (40%)
- Spikes at: 100, 200, 500, 1000
- Heavy right tail
- Not geometric, not exponential

Limit Orders:

- Modal size: 100 shares (60%)
- Similar spike pattern
- Slightly lighter tail

Dirac spikes + geometric tail

Cancellations:

- 99.3% are full cancellations
- Sample uniformly from queue
- Size distribution matches posted orders

Modeling approach:

$$P(Y = k) = \begin{cases} p_r & k \in \{1, 10, 50, \\ (1 - \sum p_r) \cdot \mathrm{Geom}(k) & \text{otherwise} \end{cases}$$

12-Dimensional CHP for LOB

Event types: $\mathcal{E} = \{LO_{ask+1}, CO_{ask+1}, LO_{ask0}, CO_{ask0}, MO_{ask0}, LO_{ask-1}, \ldots\}$

Queue size dynamics:

$$q_k^{(\zeta)}(t) = \sum_{e \in \mathcal{E}_k^{(\zeta)}} \sum_{j=1}^{N^{(i(e))}(t)} \kappa_j^{(e)}$$

where:

- $\zeta \in \{\text{ask}, \text{bid}\}$
- $k \in \{-1, 0, +1\}$ (price levels relative to best)
- $\kappa_j^{(e)} \sim \Pi^{(e)}(\Theta)$: order size for event type e

Intensity formulation:

$$\lambda^{(i)}(t) = \max\left\{0, f^{(i)}(Q_t)\left(\mu^{(i)} + \sum_{j=1}^{12} \int_0^t \phi^{(j \to i)}(t-s)\, dN^{(j)}(s)\right)\right\}$$

where $f^{(i)}(Q_t)$ is time-of-day multiplier

Spread Control Mechanism

Problem: Spread can become negative if not constrained

Solution: Make in-spread order intensity depend on current spread s

$$\lambda^{(IS)}(t, s) = \lambda^{(IS)}(t) \cdot (s - 1)^{\beta}, \quad \beta > 0$$

Properties:

- If $s = 1$ (minimum spread): $\lambda^{(IS)} = 0$ (no in-spread orders possible)
- If $s = 2$: $\lambda^{(IS)} = \lambda^{(IS)}(t)$
- If $s > 2$: $\lambda^{(IS)} > \lambda^{(IS)}(t)$ (mean reversion)

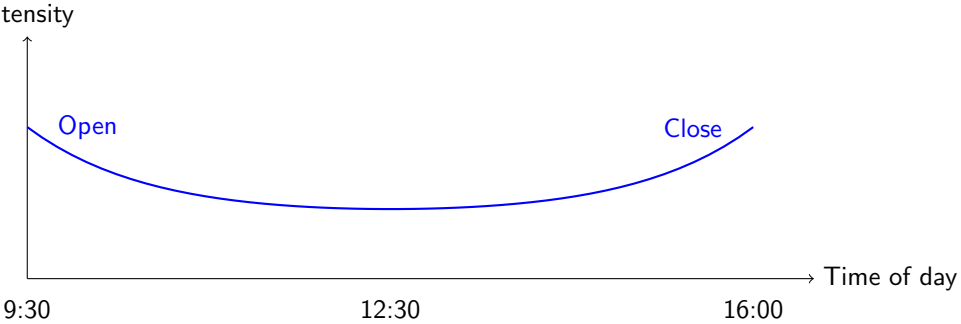
Calibration:

- Plot in-spread intensity vs. spread
- Fit power law: $\beta \approx 0.75$ for AAPL
- $R^2 = 0.85$

This ensures mechanical constraint: spread ≥ 1 tick

Time-of-Day Effects

Intraday seasonality: U-shaped order intensity pattern



Modeling approach:

- Divide day into 13 bins of 30 minutes each
- Estimate multiplier $f^{(i)}(Q_t)$ for each bin Q_t
- Separable form: $\lambda^{(i)}(Q_t, t, s) = f^{(i)}(Q_t) \times \lambda^{(i)}(t, s)$

Constraint: $\overline{f^{(i)}(Q_t)} = 1$ (average over day)

Inhibitory Cross-Excitation

Not all cross-excitations are positive!

Examples of inhibition:

1. Limit order inside spread (ask side) $\rightarrow$ Cancel at bid top
   - Aggressive ask order makes bid less attractive
   - Traders cancel their bids
2. Market order (ask side) $\rightarrow$ Market order (bid side)
   - Large buy consumes liquidity
   - Temporarily reduces sell pressure

Mathematical handling:

- Allow $\phi^{(j\to i)}(t) < 0$
- But this can make $\lambda^{(i)}(t) < 0$!
- Solution: Floor at zero

$$\lambda^{(i)}(t) = \max\left\{0, \mu^{(i)} + \sum_j \int_0^t \phi^{(j\to i)}(t-s)\, dN^{(j)}(s)\right\}$$

Full Model Summary

12D Compound Hawkes Process with:

1. **Multivariate structure:** 12 event types (6 queues $\times$ 2 sides)
2. **Order size distributions:** Calibrated for each event type
$$\kappa^{(e)} \sim \Pi^{(e)}(\Theta)$$
3. **Spread-dependent intensities:** For in-spread orders
$$\lambda^{(IS)}(t, s) = (s-1)^{\beta} \cdot \lambda^{(IS)}(t)$$
4. **Time-of-day effects:** Intraday seasonality
$$\lambda^{(i)}(Q_t, t) = f^{(i)}(Q_t) \cdot \lambda^{(i)}(t)$$
5. **Inhibitory kernels:** With non-negativity constraint
$$\lambda^{(i)}(t) = \max\{0, \cdots\}$$

This is a fully specified model of LOB dynamics!

Calibration Challenges

What needs to be estimated?

1. **Kernel functions:** $\phi^{(j \to i)}(t)$ for all $i, j \in \{1, \ldots, 12\}$
   - 144 kernels in total!
   - Need to choose functional form
2. **Baseline intensities:** $\mu^{(i)}$ for $i = 1, \ldots, 12$
3. **Time-of-day multipliers:** $f^{(i)}(Q_t)$ for 13 bins
4. **Spread parameter:** β
5. **Order size distributions:** $\Pi^{(e)}(\Theta)$ for each event type

Methods:

- Maximum Likelihood Estimation (MLE) - standard but computationally intensive
- Non-parametric methods - flexible, no prior assumptions

Maximum Likelihood Estimation

Likelihood for Hawkes process on $[0, T]$:

$$\mathcal{L} = \exp\left(-\sum_{i=1}^{d} \int_0^T \lambda^{(i)}(t)\, dt\right) \prod_{i=1}^{d} \prod_{t_i^k \leq T} \lambda^{(i)}(t_i^k)$$

Log-likelihood:

$$\ell = -\sum_{i=1}^{d} \int_0^T \lambda^{(i)}(t)\, dt + \sum_{i=1}^{d} \sum_{t_i^k \leq T} \log \lambda^{(i)}(t_i^k)$$

For exponential kernel: $\phi(t) = \alpha e^{-\beta t}$
The integral can be computed analytically:

$$\int_0^T \sum_{t_i < t} \alpha e^{-\beta(t - t_i)} dt = \frac{\alpha}{\beta} \left(N(T) - \sum_{i=1}^{N(T)} e^{-\beta(T - t_i)} \right)$$

Challenge: High-dimensional optimization (144 kernels!)

Non-Parametric Calibration (Kirchner 2017)

Key idea: Estimate kernels on a discrete grid without assuming functional form

Method:

1. Create time bins: $\Delta = t_n - t_{n-1}$ (e.g., 0.1 seconds)
2. Count events in each bin: $X_\Delta^{(e,n)} = N^{(e)}(t_n) - N^{(e)}(t_{n-1})$
3. Set up linear regression:

$$\mathbb{E}[X_\Delta^{(e,n)} | \mathcal{H}_{n-1}] \approx \Delta\mu^{(e)} + \Delta \sum_{j=1}^{d} \sum_{k=1}^{K} \frac{X_{\delta_k}^{(j)}}{\delta_k} \int_{t_{\min}^{(k)}}^{t_{\max}^{(k)}} \phi^{(j\to e)}(u)\, du$$

4. Solve least squares with constraints

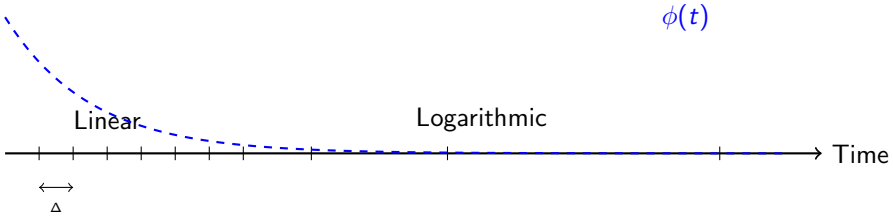
Grid choice: Linear-log spacing (following Bacry et al. 2016)

- $K/2$ linear bins for short memory
- $K/2$ logarithmic bins for long memory

Grid Construction for Slow Decay

Problem: Power-law kernels decay slowly $\Rightarrow$ need long history

Solution: Mixed linear-log grid



Grid: $\mathcal{G} = [0, \Delta, 2\Delta, \ldots, \frac{K}{2}\Delta, \frac{K}{2}\Delta m, \frac{K}{2}\Delta m^2, \ldots, T]$
where $m = \left(\frac{2T}{K\Delta}\right)^{2/K}$

Captures both short-term spikes and long-term memory

Parametric Kernel Fitting

After non-parametric estimation:
Have point estimates $\{\widehat{\Phi}(t_k)\}_{k=1}^{K}$ for each kernel

Step 2: Fit parametric function

Candidates:

1. **Exponential:** $\phi(t) = \alpha e^{-\beta t}$

$$\Phi(t) = \int_0^t \phi(s)\, ds = \frac{\alpha}{\beta}(1 - e^{-\beta t})$$

2. **Power law:** $\phi(t) = \alpha(1 + \gamma t)^{-\beta}$

$$\Phi(t) = \frac{\alpha}{\gamma(\beta - 1)}\left[1 - (1 + \gamma t)^{1-\beta}\right] \quad (\beta > 1)$$

Model selection: Akaike Information Criterion (AIC)
Result: Power law selected 100% of the time for AAPL data

Order Size Distribution Calibration

Model: Geometric base + Dirac spikes at round numbers

$$P(Y=k) = \begin{cases} p_k & k \in \{1, 10, 50, 100, 200, 500\} \\ (1 - \sum p_k) \cdot p_{\text{geom}}(k) & \text{otherwise} \end{cases}$$

where $p_{\text{geom}}(k) = (1-\theta)\theta^{k-1}$

Parameters to estimate:

- Spike probabilities: $p_1, p_{10}, p_{50}, p_{100}, p_{200}, p_{500}$
- Geometric parameter: θ

Method: Maximum Likelihood Estimation

$$\widehat{\Theta} = \arg\max_{\Theta} \sum_{i=1}^{n} \log P(Y_i|\Theta)$$

Fit separately for each event type $e \in \mathcal{E}$

Calibration Results: AAPL (2019)

Dataset: Apple Inc., NASDAQ Level 2 data, 12 months (2019)

Stock characteristics:

- Medium-tick stock
- Average spread: 1.7 ticks
- Highly liquid

Kernel selection:

- Power law chosen 100% of the time (AIC criterion)
- Mean squared error: $\approx 10^{-3}$
- Kernels stable across days (Figure in paper)

Model parsimony:

- Started with $12 \times 12 = 144$ kernels
- Threshold at $||M_{ij}|| > 0.01$
- Final model: 65 non-zero kernels

Kernel Matrix Structure

Norm matrix M reveals interaction structure:

	LO_{+1}	CO_{+1}	LO_0	CO_0	MO_0	LO_{-1}
LO_{+1}	0.45	0.12	0.08	0.03	0.01	-0.15
CO_{+1}	0.09	0.38	0.05	0.11	0.02	-0.05
LO_0	0.11	0.06	0.52	0.14	0.08	-0.22
CO_0	0.03	0.09	0.12	0.41	0.06	-0.08
MO_0	0.02	0.04	0.15	0.09	0.33	-0.03
LO_{-1}	-0.18	-0.06	-0.25	-0.10	-0.05	0.48

Observations:

- Strong diagonal: Self-excitation dominates
- Negative off-diagonal: In-spread orders inhibit opposite side
- Near-diagonal structure: Local interactions strongest

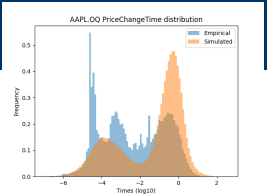
Spectral radius: $\rho(\mathbf{M}) = 0.87 < 1$ ✓ (stable)

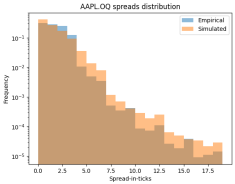
This partial matrix shows ask side only; full is 12×12

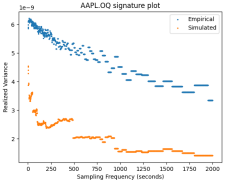
Quality of Fit: Stylized Facts

Test against multiple empirical properties:

1. **Inter-arrival time distributions**
   - Heavy left tail (clustering) ✓
   - Heavy right tail (long gaps) ✓
   - Two peaks visible ✓
2. **Spread distribution**
   - Matches empirical closely
   - Mean reversion captured
   - Spread control mechanism works
3. **Volatility signature plot**
   - Correct scale and shape
   - Microstructure noise at high frequency
   - Convergence to true volatility



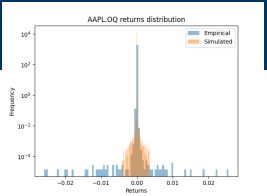


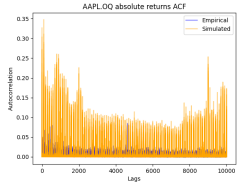


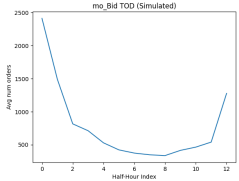
Quality of Fit: Returns and Autocorrelation

4. **Returns distribution**
   - Heavy tails (power law) ✓
   - Fat tails on log scale
   - Matches empirical well
5. **Absolute returns autocorrelation**
   - Long memory in volatility
   - ACF decays slowly
   - Clustering effect captured
6. **Time-of-day effects**
   - U-shaped pattern replicated
   - All 12 event types
   - Matches empirical seasonality

Model successfully replicates major stylized facts!







Comparison: Compound Hawkes vs Poisson

Baseline: 12D Compound Poisson Process

- Same order size distributions
- Same spread control
- Same time-of-day effects
- But: *no feedback* (independent arrivals)

Results:

Metric	Hawkes	Poisson
Inter-arrival fit	Good	Poor (missing left tail)
Spread distribution	Good	Unrealistic (too wide)
Volatility signature	Good	Flat (wrong)
Clustering	Yes	No
Price change times	Good	Too fast

Self-excitation is essential for realism!

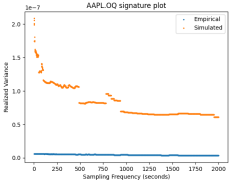
Importance of Order Size Modeling

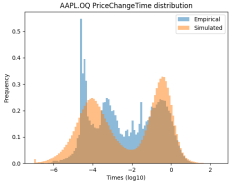
Ablation study: Hawkes with *constant* order sizes (median)

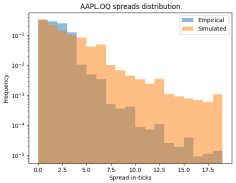
Problems observed:

1. **Volatility overestimated by $\sim 100\times$**
   - Every event has same impact
   - Queue depletion happens too frequently
2. **Price changes too fast**
   - No variation in order sizes
   - Left tail of price change times wrong
3. **Spreads too wide**
   - Excessive queue depletions
   - Mean reversion insufficient

Order size distribution is crucial for realistic dynamics!







Market Impact Study

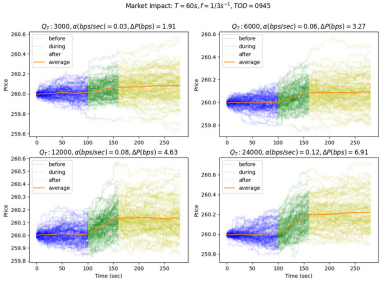
Setup: External agent executes meta-order using TWAP strategy

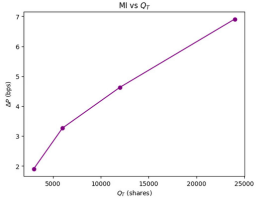
Parameters varied:

- Total quantity: $Q_T \in \{3000, 6000, 12000, 24000\}$ shares
- Duration: $T \in \{15, 30, 60, 120\}$ seconds
- Frequency: $f \in \{1, 1/3, 1/12, 1/20\}$ orders/second
- Start time: Varied throughout day

Key findings:

1. **Concave impact:** $\Delta P \propto Q_T^{\alpha}$ with $\alpha < 1$ ✓
2. **Time-of-day effect:** Higher impact at open/close
3. **Frequency effect:** More trades $\Rightarrow$ higher impact





Model replicates empirically observed market impact!

Summary

What we covered:

1. **Why EMH fails:** Clustering, feedback, market impact
2. **LOB mechanics:** Orders, queues, price formation
3. **Hawkes process theory:** Self-excitation, multivariate structure
4. **Code project:** Simplified pipeline, hands-on experience
5. **Compound Hawkes:** Adding order sizes to the model
6. **Calibration:** Non-parametric methods, MLE
7. **Empirical results:** Stylized facts, comparisons, market impact

Key message: Self-exciting point processes capture endogenous market dynamics that traditional models miss

Further Reading

Foundational papers:

- Hawkes (1971): "Spectra of some self-exciting and mutually exciting point processes"
- Bacry et al. (2015): "Hawkes processes in finance" (comprehensive review)
- Kirchner (2017): "An estimation procedure for the Hawkes process"

LOB modeling:

- Cont et al. (2010): "Statistical modeling of high-frequency financial data"
- Abergel et al. (2016): "Limit Order Books" (textbook)
- Lu & Abergel (2018): "High-dimensional Hawkes processes for limit order books"

Extensions:

- State-dependent Hawkes: Morariu-Patrichi & Pakkanen (2022)
- Neural Hawkes: Mei & Eisner (2017)

Open Questions in This Field

Active research directions:

1. **Small-tick stocks:**
   - Top-of-book models insufficient
   - Need deeper levels
   - How to maintain parsimony?
2. **State dependence:**
   - Should ϕ depend on queue sizes?
   - Spread? Order flow imbalance?
3. **Inference at scale:**
   - High-dimensional models (12D, 20D, ...)
   - Computational challenges
4. **Theoretical properties:**
   - Scaling limits for compound Hawkes
   - Connection to price diffusion

Thank You!

Questions?

Contact: konark.jain.23@ucl.ac.uk / konark145@gmail.com
Code: GitHub

Good luck with your mini-project!

Appendix: Thinning Algorithm

Simulation of Hawkes process:

Ogata's thinning algorithm:

```
Initialize t ← 0, λ* ← μ
while t < T do
    Sample u ~ Unif(0, 1)
    t ← t − (1/λ*) log(u)
    if t > T then
        break
    end if
    Sample v ~ Unif(0, 1)
    Compute λ(t)
    if v ≤ λ(t)/λ* then
        Accept event at time t
        Update λ* ← λ(t) + α
    else
        Reject (continue)
    end if
```

Appendix: Likelihood Computation

Log-likelihood for exponential kernel:

$$
\begin{aligned}
\ell &= -\int_0^T \lambda(t)\,dt + \sum_{i=1}^{n} \log \lambda(t_i) \\
&= -\int_0^T \left(\mu + \alpha \sum_{t_j < t} e^{-\beta(t-t_j)} \right) dt + \sum_{i=1}^{n} \log \lambda(t_i) \\
&= -\mu T - \alpha \sum_{j=1}^{n} \int_{t_j}^{T} e^{-\beta(t-t_j)} dt + \sum_{i=1}^{n} \log \lambda(t_i) \\
&= -\mu T - \frac{\alpha}{\beta} \sum_{j=1}^{n} (1 - e^{-\beta(T-t_j)}) + \sum_{i=1}^{n} \log \lambda(t_i)
\end{aligned}
$$

Complexity: $O(n)$ per evaluation using recursion for $\lambda(t_i)$